

E-Commerce Customer Segmentation

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July 2026

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Chapter 1

Introduction

1.1 Introduction

Many businesses use Customer Segmentation as an important technique to understand consumer behaviour. Using this technique, they can treat and study these customers in smaller segments based on shared characteristics. By doing so, they get the opportunity to design personalized marketing campaigns, improve customer satisfaction, and optimize resource allocation.

Machine learning is an up-and-coming tool that has become invaluable for Customer Segmentation. As it allows businesses to identify hidden patterns without any help from predefined labels, it has become increasingly popular. One of the most sought-after ML methods is **K-Means Clustering**.

K-Means Clustering has been used in this project to analyze consumer data and study demographic and spending-related attributes such as age, annual income, and spending score. I have also made use of other methods like data processing, EDA, feature scaling, clustering, and visualization of customer segments. Finally, an interactive dashboard has been developed using Streamlit that allows users to explore the dataset and clustering results in a user-friendly manner.

1.2 Industry Background

Leading companies such as Amazon, Flipkart, Myntra, Walmart, Alibaba, and eBay use customer segmentation to personalize shopping experiences and optimize marketing campaigns. Applications include:

- Personalized product recommendations
- Targeted advertisements
- Customer loyalty programs
- Market basket analysis
- Customer lifetime value prediction
- Sales optimization

Customer segmentation is one of the most important applications of Data Science in the retail industry.

1.3 What's the problem?

E-commerce companies serve millions of customers with different purchasing behaviors. Without customer segmentation, businesses struggle to provide personalized experiences and targeted marketing.

The objective of this project is to classify customers into meaningful groups using data analysis and machine learning techniques.

1.4 Objectives

- To study customer purchasing behaviour using demographic and spending-related data.
- To preprocess and clean the dataset for analysis.
- To perform Exploratory Data Analysis (EDA) to understand customer characteristics.
- To implement the K-Means clustering algorithm for customer segmentation.
- To determine the optimal number of clusters using the Elbow Method.
- To evaluate clustering quality using the Silhouette Score.
- To visualize customer segments using appropriate charts and plots.
- To develop an interactive Streamlit dashboard for presenting the results.
- To generate business insights based on the identified customer segments.

Chapter 2

Dataset Description

2.1 Source of the Data

For this project, I have used a dataset available on Kaggle titled [store_customers.csv](#). This is a dataset commonly used for Customer Segmentation and clustering which makes it a good fit for the project. It includes all important demographic and spending-related information of the customers at the mall. While inspecting the dataset, I found the following columns (Refer to Figure 2.1): As can

```
[38]:  
df.columns  
  
[38]:  
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',  
      'Spending Score (1-100)'],  
      dtype='object')
```

Figure 2.1: Columns in the Dataset

be seen, this information is sufficient to analyze customer behavior and identify distinct customer groups using machine learning techniques.

The dataset includes 1000 customer records, which have all the information corresponding to the aforementioned titles. These attributes provide valuable insights into customer purchasing behavior and are appropriate for applying clustering algorithms like K-Means.

2.2 Dataset Overview

Before doing the analysis, the structure and other important qualities of the dataset were also examined properly. This included identifying missing values and verifying the data types of each feature. Data preprocessing was performed to ensure consistency and improve the quality of the dataset before applying machine learning techniques.

The final dataset contains five attributes, including one unique identifier, one categorical feature,

```
[37]:
df.info

[37]:
<bound method DataFrame.info of
ng Score (1-100)
0      1000      M  39.0      59.9      58.0
1      1001      M  34.0      48.4      37.0
2      1002      F  40.0      70.5      26.0
3      1003      F  47.0      81.1      30.0
4      1004      F  33.0      42.1      58.0
..      ...      ...      ...      ...
995     1995      M  80.0     133.3      1.0
996     1996      M  44.0      82.6     40.0
997     1997      F  46.0      67.7     26.0
998     1998      F  28.0      45.7     59.0
999     1999      M  67.0     107.8      1.0

[1000 rows x 5 columns]>
```

Figure 2.2: Information of the Dataset

```
[65]:
df.isnull().sum()

[65]:
CustomerID      0
Gender          3
Age            6
Annual_Income   4
Spending_Score  6
dtype: int64
```

Figure 2.3: Identifying missing values in the Dataset

and three numerical features. These variables provide the necessary information for performing customer segmentation.

2.3 Features

Features of the Dataset		
Feature	Description	Data Type
Customer ID	Unique identification number assigned to each customer	Integer
Gender	Gender of the Customer	Categorical
Age	Age of the customer in years	Numerical
Annual Income (k\$)	Annual income of the customer (in thousands of dollars)	Numerical
Spending Score (1-100)	Score assigned based on customer spending behaviour	Numerical

2.4 Importance

The table helps us understand how each value is of significance for this project. We have both demographic and behavioural information. Variables such as age, annual income, and spending score allow the clustering algorithm to identify customers with similar purchasing patterns. The insights obtained from these customer groups can assist businesses in designing targeted marketing strategies, improving customer retention, and enhancing overall decision-making.

Furthermore, the dataset provides an appropriate balance between simplicity and practical relevance, making it suitable for demonstrating the application of K-Means clustering in a real-world business context.

Chapter 3

Methodology

3.1 Data Collection

First and foremost, I had to import the data from the dataset into my Python notebook to start working on it. To do so, I used the pandas library. The dataset was stored in CSV (Comma-Separated Values) format and loaded into a DataFrame for further preprocessing and analysis. After loading the dataset, its dimensions, column names, and data types were examined to obtain an initial understanding of the data structure.

```
[4]:
df = pd.read_csv("../datasets/store_customers.csv")
df.head()
```

[4]:

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1000	M	39.0	59.9	58.0
1	1001	M	34.0	48.4	37.0
2	1002	F	40.0	70.5	26.0
3	1003	F	47.0	81.1	30.0
4	1004	F	33.0	42.1	58.0

Figure 3.1: Preview of the Customer Dataset

3.2 Data Cleaning

3.2.1 Inspection

Before preprocessing, the dataset was inspected to identify missing values, duplicate records, and inconsistencies in the data.

Functions such as `head()`, `info()`, `describe()`, `columns`, and `dtypes` were used to understand the characteristics of the dataset. This inspection ensured that appropriate preprocessing steps could be applied before performing machine learning analysis.

```
[13]:
```

```
df.describe()
```

```
[13]:
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	1000.000000	994.000000	996.000000	994.000000
mean	1499.500000	38.935614	57.149096	42.645875
std	288.819436	13.399880	28.628506	20.101589
min	1000.000000	18.000000	15.000000	1.000000
25%	1249.750000	30.000000	34.975000	31.000000
50%	1499.500000	36.000000	49.000000	47.000000
75%	1749.250000	44.000000	79.400000	57.000000
max	1999.000000	80.000000	144.100000	92.000000

Figure 3.2: Description of the Dataset

```
[16]:
```

```
df.dtypes
```

```
[16]:
```

CustomerID	int64
Gender	object
Age	float64
Annual Income (k\$)	float64
Spending Score (1-100)	float64
dtype:	object

Figure 3.3: Checking the Data Type

3.2.2 Handling Missing Values

During inspection, missing values were identified in several columns (Refer to Figure 2.3). Since the K-Means clustering algorithm cannot process missing values directly, appropriate preprocessing was performed to remove incomplete records before training the model. This ensured that the dataset used for clustering contained valid observations without null values, improving the reliability of the clustering results (Refer to Figure 3.4).

3.2.3 Removing Duplicates

Duplicate records were examined to ensure that repeated customer entries did not influence the clustering process. Duplicate observations, if present, were removed using the Pandas `drop_duplicates()` function. Removing duplicate records improves data quality and prevents certain observations from having an unintended influence on the clustering algorithm (Refer to Figure 3.5).

3.2.4 Selecting the right features

I chose the Annual Income and Spending Score features because they best represent customer purchasing behaviour and provide meaningful separation between customer groups. The CustomerID column was excluded since it serves only as a unique identifier, while Gender and Age were analyzed during exploratory data analysis but were not used as clustering features in the K-Means model (Refer to Figure 3.6).

```
[9]:
df.dropna()
```

```
[9]:
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1000	M	39.0	59.9	58.0
1	1001	M	34.0	48.4	37.0
2	1002	F	40.0	70.5	26.0
3	1003	F	47.0	81.1	30.0
4	1004	F	33.0	42.1	58.0
...	...	...	...	...	...
995	1995	M	80.0	133.3	1.0
996	1996	M	44.0	82.6	40.0
997	1997	F	46.0	67.7	26.0
998	1998	F	28.0	45.7	59.0
999	1999	M	67.0	107.8	1.0

982 rows × 5 columns

Figure 3.4: Dropping the missing values

```
[10]:
df.duplicated().sum()
```

```
[10]:
np.int64(0)
```

```
[17]:
df = df.drop_duplicates()
```

Figure 3.5: Dealing with the Duplicates

```
[70]:
df.rename(columns={
    "Annual Income (k$)": "Annual_Income",
    "Spending Score (1-100)": "Spending_Score"
}, inplace=True)
```

```
[71]:
df.columns
```

```
[71]:
Index(['CustomerID', 'Gender', 'Age', 'Annual_Income', 'Spending_Score'], dtype='object')
```

```
[72]:
df.to_csv("../datasets/Mall_Customers_Clean.csv", index=False)
```

Figure 3.6: Selecting Features

Chapter 4

Exploratory Data Analysis (EDA)

4.1 Introduction to Exploratory Data Analysis

Exploratory Data Analysis (EDA) is an essential step in the data science workflow that involves examining and visualizing the dataset to understand its underlying structure, identify patterns, detect anomalies, and summarize the main characteristics of the data.

In this project, EDA was performed using statistical summaries and graphical visualizations to gain insights into customer demographics and purchasing behavior before applying the K-Means clustering algorithm. The analysis helped identify trends, potential outliers, and relationships among the variables, providing a strong foundation for the clustering process.

4.2 Gender Distribution

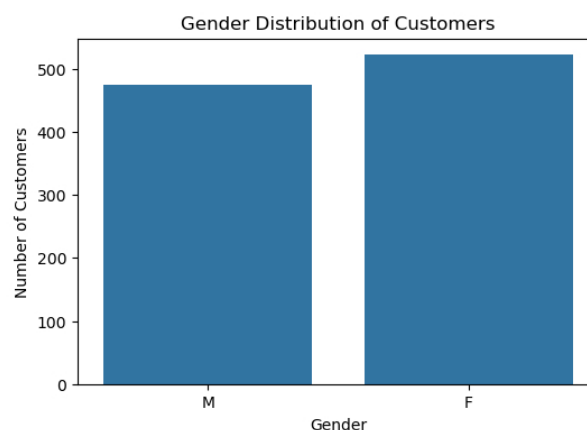


Figure 4.1: Gender Distribution Chart

Looking at the Gender Distribution Chart helps us understand the number of male and female

customers. As can be seen, there is but little difference in their numbers, with female customers leading by a few digits.

Information about customers from both genders allows the clustering algorithm to analyze purchasing behavior across a diverse customer base. However, gender was used only for exploratory analysis and was not included as a feature in the K-Means clustering model.

4.3 Age Distribution

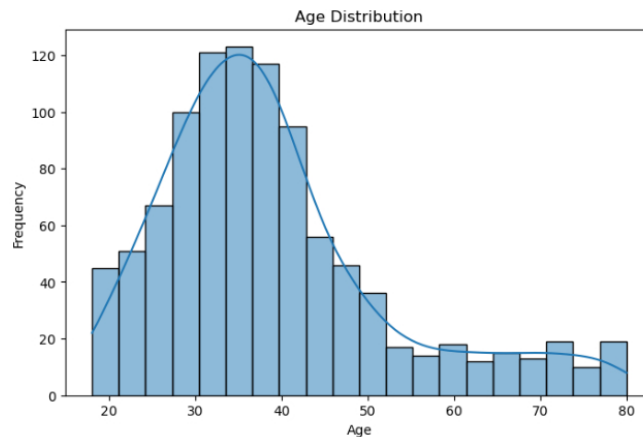


Figure 4.2: Age Distribution Chart

I have used a [histplot](#) to show that customers belong to a wide range of age groups. Most customers are concentrated within the young adult and middle-aged categories, while comparatively fewer customers belong to the higher age groups. The distribution indicates diversity in customer demographics and highlights the presence of a few older customers. I have studied the small number of elderly customers present in a boxplot later 4.5.

These observations provide useful context for understanding customer purchasing behavior. For instance, when coming up with mall policies and plans, we will have to keep the larger group in mind. But at the same time, since we have data on older customers, we will have to make sure that they don't get inconvenienced.

4.4 Annual Income Distribution

The annual income distribution demonstrates variation in customer income levels across the dataset. Most customers fall within the middle-income range, while fewer customers are observed at the extreme lower and higher income levels. This helps us identify that most of our consumer base is part of the middle class. This information would come in handy when we make plans in the future.

Understanding income distribution is important because annual income is one of the primary variables used for customer segmentation through K-Means clustering.

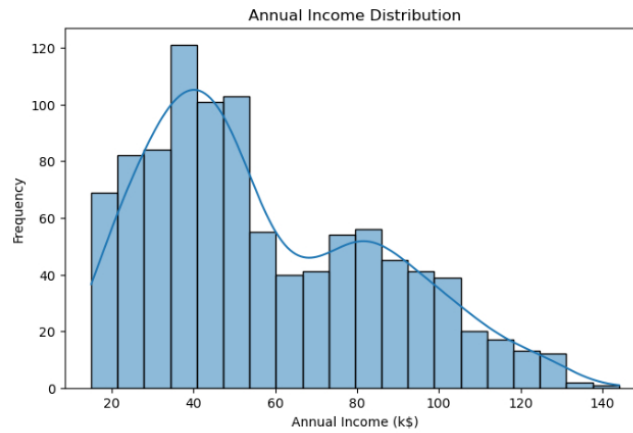


Figure 4.3: Annual Income Distribution Chart

4.5 Spending Score Distribution

The spending score represents the purchasing behavior of customers on a scale from 1 to 100. The distribution indicates that customers exhibit varying spending habits, ranging from low to high spending levels. This variation provides meaningful information for identifying distinct customer segments during clustering.

There are two peaks in the plot - people are either spending very little, or they are spending a lot. I analyse this further in later sections.

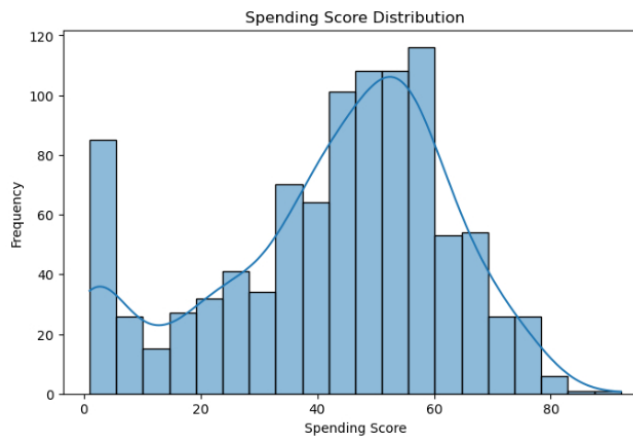


Figure 4.4: Spending Score Distribution Chart

4.6 Boxplot Analysis

Boxplots are used to identify the spread of the numerical variables and detect potential outliers. As

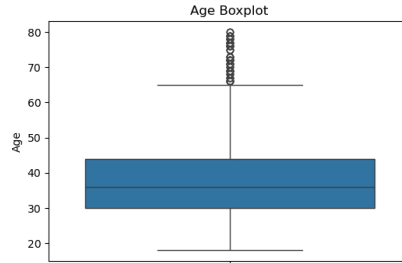


Figure 4.5: Boxplot for Age

can be seen here, the outliers in the case of age appear to be the elderly customers. This means that out of the total number, most of them are part of the younger population, while a few are older than 60.

4.7 Correlation Analysis

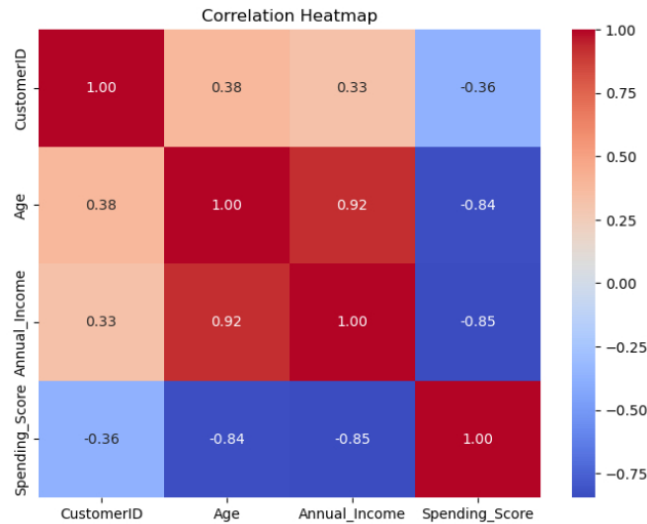


Figure 4.6: Correlation Heatmap

The correlation heatmap illustrates the relationships among the numerical variables in the dataset. The analysis indicates that no strong linear correlation exists between annual income, spending score, and age. This suggests that each variable contributes unique information to the

customer segmentation process. This supports the use of clustering techniques to identify patterns that are not immediately apparent through pairwise correlations.

4.8 Some more analysis

Using the spending score chart, I noticed that people are either spending a lot of money or they aren't. I wanted to investigate this relationship further by comparing the spending score with age, gender and income to see if anything comes up.

4.8.1 Age

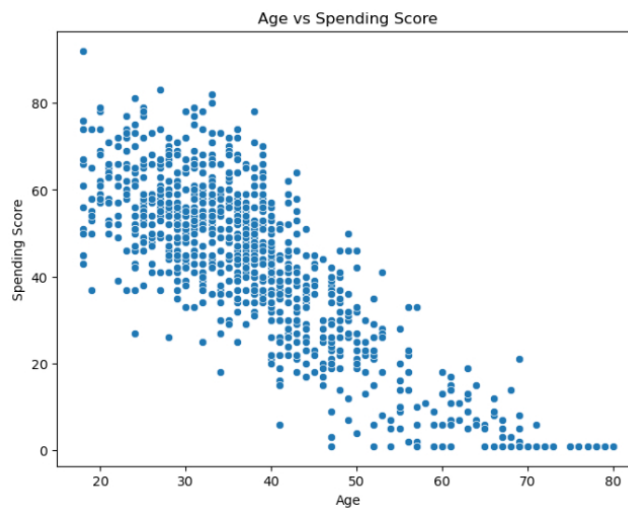


Figure 4.7: Age Vs Spending Score

I chose a [scatterplot](#) for this comparison, and it proved to be very helpful because there is a clear relationship between the two values. The slope is negative - as people are growing older, their spending score is decreasing! There are very few older customers with high spending scores. There is some variability in the younger population, but in general, it can be inferred that age may influence purchasing behaviour.

4.8.2 Gender

Since gender is a categorical value, I went with a [boxplot](#) for analysis. However, unlike the previous case, both boxplots look very similar, and the medians are almost identical. This means that there is no obvious difference in spending behaviour between male and female customers for this particular place (the mall).

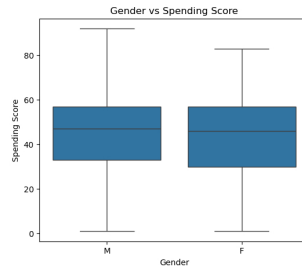


Figure 4.8: Gender Vs Spending Score



Figure 4.9: Annual Income Vs Spending Score

4.8.3 Annual Income

In this case as well, there is a negative trend. In general, as annual income increases, spending scores tend to decrease. However, the plot is important for one more reason. Several dense regions of points are visible, suggesting natural customer groupings, which provide strong justification for applying the K-Means clustering algorithm to identify meaningful customer segments.

4.9 Pearson correlation coefficients

Since I observed a noticeable negative relationship in the first and third plots, I calculated the **Pearson correlation coefficients** for the two cases: the Age and the Spending Score, and the Annual Income and the Spending Score.

This coefficient tells us how strongly two variables are linearly related, and ranges from -1 to +1. You can refer to the table for better understanding.

Pearson Correlation Coefficient Chart		
Correlation (r)	Meaning	Interpretation
+1.0	Perfect positive	As one variable increases, the other always increases.
+0.7 to +0.9	Strong positive	The variables generally increase together.
+0.3 to +0.7	Moderate positive	A noticeable positive relationship exists.
0	No linear relationship	One variable does not linearly predict the other.
-0.3 to -0.7	Moderate negative	As one variable increases, the other tends to decrease.
-0.7 to -0.9	Strong negative	A clear inverse relationship exists.
-1.0	Perfect negative	As one variable increases, the other always decreases.

Now, for the case of Age and Spending Score, the value -0.835 is very close to -1, which indicates a strong negative linear relationship. This confirms our understanding that as Age increases, Spending Score tends to decrease.

The Coefficient for Annual Income is -0.846073, which is even stronger than the other. This means that as Annual Income increases, Spending Score generally decreases.

[53]:		
<code>df[["Age", "Spending_Score"]].corr()</code>		
[53]:		
	Age	Spending_Score
Age	1.000000	-0.835186
Spending_Score	-0.835186	1.000000
[54]:		
<code>df[["Annual_Income", "Spending_Score"]].corr()</code>		
[54]:		
	Annual_Income	Spending_Score
Annual_Income	1.000000	-0.846073
Spending_Score	-0.846073	1.000000

Figure 4.10: Pearson correlation coefficients

Chapter 5

Customer Segmentation using K-Means Clustering

5.1 Introduction to K-Means Clustering

K-Means is an unsupervised machine learning algorithm widely used for clustering tasks. Unlike supervised learning algorithms, K-Means does not require labeled data. Instead, it partitions observations into a predefined number of clusters based on the similarity of their features. The algorithm aims to minimize the distance between data points and the centroid of the cluster to which they belong. In this project, K-Means clustering was employed to identify groups of customers with similar purchasing behaviour based on their annual income and spending score.

5.2 Feature Selection

Selecting appropriate features is a critical step in the clustering process. For this project, **Annual Income** and **Spending Score** were chosen as the clustering variables because they directly represent customers' purchasing capacity and spending behaviour.

Selected Feature	Reason for Selection
Annual Income (k\$)	Represents the purchasing capacity of customers and serves as a key indicator for customer segmentation.
Spending Score (1–100)	Represents the spending behaviour of customers and reflects their purchasing habits.

5.3 Feature Scaling

Since K-Means is a distance-based algorithm, feature scaling was performed before clustering. Standardization was applied using the [StandardScaler](#) technique, which transforms each feature to have a mean of zero and a standard deviation of one. This prevents variables with larger numerical values from dominating the clustering process and ensures that both features contribute equally to the calculation of distances.

```
[103]:
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Figure 5.1: Feature Scaling

5.4 Determining the Optimal Number of Clusters

Selecting an appropriate number of clusters is essential for effective segmentation. The Elbow Method was used to determine the optimal value of k by plotting the **Within-Cluster Sum of Squares (WCSS)** against different numbers of clusters. The graph showed a noticeable bend at

$$k = 4$$

indicating that increasing the number of clusters beyond this point resulted in only marginal improvements. Therefore, four clusters were selected for the K-Means model.

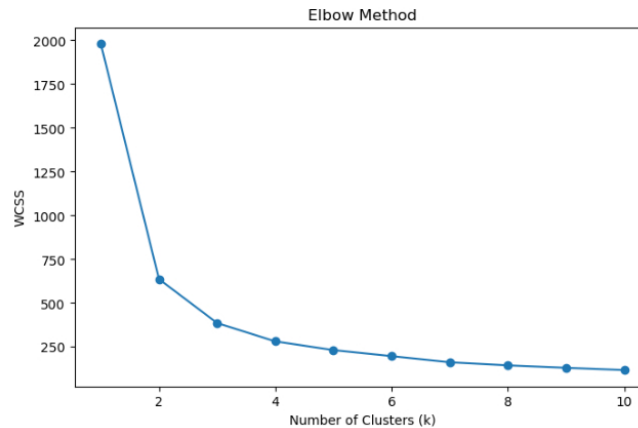


Figure 5.2: Elbow Method

5.5 Modal Training

After selecting the optimal number of clusters, the K-Means algorithm was trained using the standardized annual income and spending score features. Each customer was assigned to one of four clusters based on similarity to the corresponding cluster centroid. The trained model produced customer groups that were later visualized and analyzed to identify distinct purchasing patterns.

5.6 Cluster Visualization

The clustered scatter plot illustrates the customer segments identified by the K-Means algorithm. Each colour represents a unique customer segment, while the black markers indicate the cluster

centroids. The visualization demonstrates that the algorithm successfully separated customers into distinct groups based on annual income and spending behaviour. The centroids represent the average characteristics of each cluster and serve as the reference point for assigning customers to their respective segments.

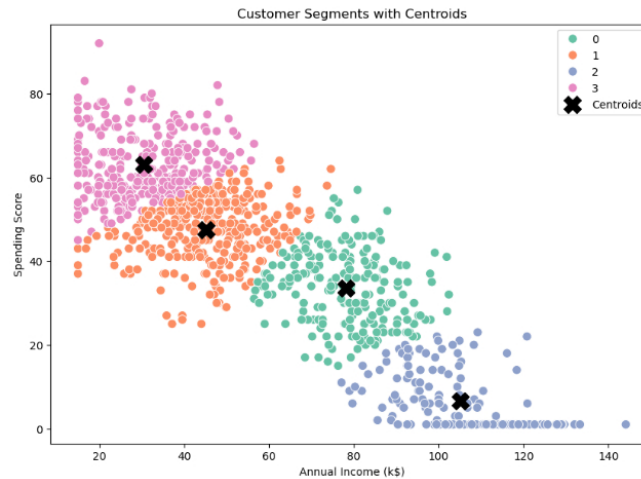


Figure 5.3: Cluster Visualization with Centroids

5.7 Cluster Analysis

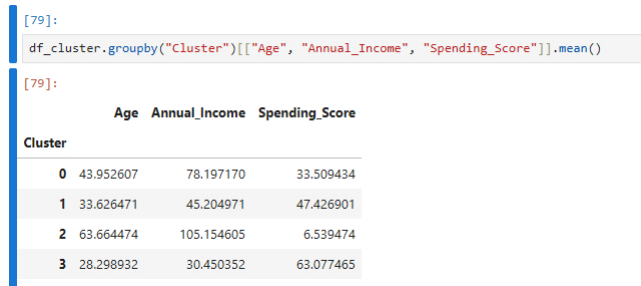


Figure 5.4: Cluster Analysis

5.7.1 Cluster 0

Looking at the table tells us that this cluster consists of middle-aged customers with relatively high incomes but below-average spending scores. Despite having good purchasing power, they spend conservatively.

5.7.2 Cluster 1

These are young to middle-aged customers with moderate incomes and moderate spending habits. They represent a stable customer group with balanced purchasing behaviour.

5.7.3 Cluster 2

This segment consists of older, high-income customers who spend very little. They have the highest average income but the lowest spending score, indicating untapped purchasing potential.

5.7.4 Cluster 3

These customers are the youngest group with the lowest average income but the highest spending score. They appear to be highly engaged shoppers despite having relatively limited purchasing power.

5.8 Model Evaluation

The performance of the clustering model was evaluated using the **Silhouette Score**, which measures how well each data point fits within its assigned cluster compared to neighbouring clusters. The model achieved a **Silhouette Score of 0.3948**, indicating a reasonable level of cluster separation and cohesion. Although the clusters are not perfectly distinct, the score suggests that the K-Means algorithm successfully identified meaningful customer groups within the dataset.

5.9 Dashboard Implementation

To improve the accessibility and visualization of the customer segmentation results, an interactive dashboard was developed using Streamlit. The dashboard allows users to explore the dataset, visualize exploratory data analysis, inspect customer clusters, and review business insights through an intuitive graphical interface. This provides a user-friendly method for presenting analytical results without requiring direct interaction with the Python code.

Chapter 6

Conclusion and Business Insights

6.1 Conclusion

This project successfully demonstrated the application of K-Means clustering for customer segmentation using demographic and spending-related information obtained from a customer dataset. The primary objective was to identify meaningful customer groups that could support data-driven business decision-making and targeted marketing strategies.

The project began with data preprocessing, including handling missing values, removing duplicate records, verifying data types, and preparing the dataset for analysis. Exploratory Data Analysis (EDA) was then performed to examine the characteristics of the data and identify relationships among customer attributes. The analysis revealed that both age and annual income exhibited strong negative correlations with spending score, while gender showed no significant influence on customer spending behaviour.

K-Means clustering was applied using Annual Income and Spending Score as the clustering variables. The Elbow Method was used to determine the optimal number of clusters, and the resulting customer segments were analyzed based on their demographic and behavioural characteristics. Each cluster represented a distinct customer profile, enabling meaningful interpretation of purchasing patterns.

To enhance usability, an interactive dashboard was developed using Streamlit. The dashboard provides an intuitive interface for exploring the dataset, visualizations, clustering results, and business insights, making the analytical findings more accessible to end users.

Overall, the project demonstrates how machine learning techniques can transform customer data into actionable insights that support strategic business decisions and improve customer relationship management.

6.2 Business Insights

- The K-Means algorithm successfully divided customers into four distinct segments based on annual income and spending behaviour.
- Younger customers generally exhibited higher spending scores than older customers.
- Customers with high annual incomes did not necessarily spend more, indicating untapped market potential.

- Gender did not show a significant influence on spending behaviour.
- Customer segmentation enables businesses to understand purchasing patterns and tailor marketing strategies accordingly.

6.3 Business Recommendations

Customer Segment	Recommended Strategy
Cluster 0	Encourage higher spending through personalized promotions, premium membership programs, and targeted discounts.
Cluster 1	Maintain customer engagement using loyalty programs, seasonal offers, and personalized product recommendations.
Cluster 2	Develop exclusive offers, premium experiences, and individualized marketing campaigns to increase purchasing activity.
Cluster 3	Reward loyal customers with discounts, referral programs, and products tailored to younger consumers.

6.4 Limitations

Although the proposed clustering model produced meaningful customer segments, several limitations should be acknowledged. The analysis was performed using a limited number of customer attributes, including age, gender, annual income, and spending score. Incorporating additional information such as **purchase history**, **customer location**, **product preferences**, or **transaction frequency** could provide more detailed customer profiles. Furthermore, K-Means requires the number of clusters to be specified in advance and assumes that clusters are relatively compact, which may not always reflect real-world customer behaviour.

6.5 Future Scope

- Apply alternative clustering algorithms such as DBSCAN, Agglomerative Clustering, or Gaussian Mixture Models and compare their performance.
- Incorporate additional customer attributes, including purchase history, browsing behaviour, and loyalty information.
- Develop a real-time customer segmentation system that updates as new customer data becomes available.
- Integrate the dashboard with a live business database for interactive decision support.
- Enhance the dashboard by adding predictive analytics and personalized recommendation features.

The integration of machine learning with an interactive Streamlit dashboard demonstrates how analytical models can be transformed into practical decision-support tools, enabling businesses to explore customer behaviour and make informed, data-driven marketing decisions.